

LIRUI WANG

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EDUCATION

University of Washington, Seattle, WA

Sep 2016 - Present

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

GPA: 3.97/4.00, Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

Joint Motion and Grasp Planning

March 2019 - Present

- Built a neural net that synthesizes grasp poses based on 3D geometry.
- Prototyped RL, IL, and MPC for manipulators in PyBullet environment.
- Implemented CHOMP Planner and introduced an integrate planner that optimizes manipulation trajectory with online grasp synthesis and selection.
- Proposed an online learning framework and a grasp synthesis approach to generate grasp and select grasp online.

Pose Tracking with the Deep Iterative Matching Network

June 2018 - March 2019

- Tested DeepIM on YCB datasets and achieved robust results trained on synthetic data only.
- Improved network training stability, pose distribution and occlusion generation.
- Extended DeepIM to real-time tracking for occluded scenes and Augmented Reality applications.
- Built a OpenGL renderer and kinematics tree for vision system application on robotics.

Unsupervised Flow and Depth Methods Study

March 2018 - June 2018

- Investigated deep learning methods for 3D geometry tasks such as depths, flow, and egomotion.
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a manipulation setting.

Baxter Hand-Eye Coordination

Aug 2017 - March 2018

- Worked on a manipulation task with PoseCNN, which enables Baxter to move daily objects.
- Studied camera models, hand-eye calibration with depth camera, robot kinematics, 3D geometry and ROS packages for tracking and control.

WORK EXPERIENCE

Seattle, WA

Sep 2019 - Present

Computer Vision Teaching Assistant, UW CSE 455

Shenzhen, China

Jun 2017 - Aug 2017

Robotics Engineer Intern, DJI Technology

- Worked on autonomous robot battle which uses vehicles with TX1 onboard and DJI Manifold drones with Guidance
- Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate
- Applied SLAM to mapping and localize, and built a top-level state machine to strategize and navigate the robot

LEADERSHIP AND INVOLVEMENT

Seattle, WA

Sep 2016 - Feb 2018

Computer Vision Lead

Advanced Robotics at the University of Washington

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI.
- Performed system integration of controls, navigation and serial communication.

MISCELLANEOUS PROJECTS

- **Train SoundScape Simulator:** Designed a sound processing pipeline from scratch for modelling the acoustics of a train.
- **Sheet Music Editor:** Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded 1st place out of 35 teams in Seattle.
- **Embedded Drowsiness Detection System:** Built an embedded system on Raspberry Pi that that simulates a real-time drowsiness detection system based on human face detection.
- **Survey and Empirical Evaluation of Attacks and Defense for Robust Classification:** Investigate example reweighting and adversarial training techniques to robustify a fossil classification task and survey recent papers about attacks and defense of adversarial examples for neural networks.
- **Continuous Trajectory Optimization with Machine Learning Techniques:** Summarize the formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space and as probabilistic inference using Gaussian Process.

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*

PUBLICATIONS

- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
Lirui Wang, Yu Xiang, Dieter Fox
In Submission of The International Conference on Automated Planning and Scheduling (ICAPS 2020)

COURSES

- **Computer Science:** Data Structure and Parallelism, Algorithms, Computer Vision, Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Robotics, Robustness in Machine Learning (Grad)
- **Electrical Engineering:** Digital Signal Processing, Control Theory, Control Capstones, Embedded System
- **Math:** Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Topology, Convex Optimization and Analysis (Grad), High Dimensional Statistics and Statistical Learning (Grad), Statistical Learning: Modeling, Prediction, And Computing (Grad)
- **English:** Technical Writing, Advanced Technical Writing in Electrical Engineering

SKILLS

- **Systems:** ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32
- **Languages:** Python, C++/C, Matlab, Java
- **Tools:** Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt, Bullet, VTK,
- **Hardware:** 3D Printing, Laser Cutting, Machining, Soldering

EXTRACURRICULARS

- **Mens Pickup Soccer Team:** Played regularly at pickup soccer and joined the intramural soccer competition.
- **Math Club:** Joined the meetings to solve problems with other students in the group.